SAI SPOORTHY VARALA

AI ENGINEER | LLM APPLICATIONS • GENERATIVE AI • RAG • HEALTHCARE AI
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PROFESSIONAL SUMMARY

Software engineer with 4+ years of experience building end-to-end ML and generative AI systems, from data processing and model evaluation to production deployment. Built agentic LLM workflows, real-time speech-to-text pipelines, and LLM-as-a-Judge evaluation systems for automated quality assurance. Full-stack experience with Python, Java, React, REST APIs, and database schemas, deployed on AWS and Azure with CI/CD (GitHub Actions)

PROFESSIONAL EXPERIENCE

HealthXMD, Inc. **Jul 2024 - Present**
Software Developer, Level II *Los Angeles, CA*

- Designed and developed HIPAA-compliant AI clinical documentation services using Python, OpenAI GPT models, LangChain, and Flask, cutting clinicians assessment time from 40-50 to 10-15 minutes (~70% reduction)
- Engineered high-throughput GenAI data pipelines in Python that convert transcripts into structured data in ~30 seconds, using prompt design, structured outputs, and parallel batch processing, and designed the database schemas in SQL Server and MongoDB
- Established the chart-quality standard by designing a 40+ criterion evaluation rubric covering required fields, documentation rules, and care-plan requirements, and wrote the SOPs and pass/fail rules for reviewing flagged charts, which cut review time per flagged chart from ~15 to ~5 minutes
- Architected an LLM-as-a-Judge QC/QA engine that expanded chart review from manual sampling to 100% automated daily coverage, flagging missing documentation, inconsistencies, and care-plan gaps. 88% of flagged inconsistencies were resolved and 87% of flagged plan-of-care gaps were added to charts
- Developed a multimodal voice-to-chart workflow pairing a real-time speech-to-text front end with the LLM extraction pipeline and secure mobile APIs, eliminating manual note entry and producing structured charts within ~30 seconds of a visit ending
- Implemented asynchronous back-end services and microservices with REST APIs secured by HMAC request signing and JWT authentication on AWS (EC2, ECS), Docker, and GitHub Actions CI/CD, cutting deployment time from ~2 hours to under 15 minutes
- Automated a hybrid rule-based and LLM pipeline for pharmacy invoices that ingests email attachments via the Gmail API, parses ~70% with Python/regex, and falls back to OpenAI GPT models for the rest, reducing manual billing reconciliation time by ~60%

Accenture Solutions **Jul 2021 - Aug 2022**
Application Development Analyst *Hyderabad, India*

- Migrated 15 on-premises enterprise PLM applications and their data to Microsoft Azure using Azure Data Factory and Azure Database Migration Service, moving SQL Server databases to Azure SQL Managed Instance with zero data loss through automated validation and testing
- Provisioned Azure VMs, Load Balancer, Virtual Networks, and Key Vault through ARM templates and Azure DevOps pipelines, cutting environment setup from ~2 days to ~4 hours, and configured Azure Backup and Site Recovery with disaster-recovery failover validated for all 15 applications
- Monitored VM and application health with Azure Monitor and Log Analytics, and worked with engineering teams to triage and resolve 60+ production incidents across integrated modules after go-live

Techvolt **May 2020 - Dec 2020**
Machine Learning Engineer *Coimbatore, India*

- Constructed end-to-end machine learning pipelines in Python, Pandas, and Scikit-learn, from ETL ingestion of API, database, and log data through data cleaning, feature engineering, and exploratory analysis on 50K+ records, surfacing customer-behavior trends, payment patterns, and anomalies
- Trained and evaluated 7 classification and regression models (logistic and linear regression, tree-based models, SVM, KNN, Naive Bayes), tuning with cross-validation and hyperparameter search. The best model reached 93% accuracy, up from an 81% baseline
- Deployed trained models as real-time prediction services with REST APIs integrated into back-end applications, serving predictions with under 100 ms response time

DoelSoft Solutions **Jun 2019 - Nov 2019**
Software Developer Intern *Hyderabad, India*

- Built back-end services for a hotel booking platform using Java, Spring Boot, and MySQL, designing the database schema for booking and user-management features and optimizing queries with indexing, which cut room-search response time from ~2 seconds to under 500 ms
- Delivered 15+ RESTful APIs with Spring Boot for room search, reservations, and user authentication, using a layered controller-service-repository design and JUnit tests to keep business logic modular and testable

TECHNICAL SKILLS

- **Programming Languages & Libraries:** Python, SQL, Java, C#, JavaScript, TensorFlow, Scikit-learn, Pandas, React.js, TypeScript
- **AI & Generative AI:** LLMs (Anthropic Claude, OpenAI GPT), LangChain, AI Agents, Agentic Workflows, Tool Calling, RAG, Embeddings, Prompt Design & Optimization, Structured Outputs, Speech-to-Text, NLP, Document Processing
- **Machine Learning & Evaluation:** End-to-End ML, Model Training & Deployment, Model Evaluation (Precision, Recall, Cross-Validation), LLM-as-a-Judge, Evaluation Rubrics, LLM Output Validation, Data Quality Validation, Root Cause Analysis, TensorFlow, Scikit-learn, Pandas
- **LLM Evaluation & Data Quality:** LLM-as-a-Judge, Judge Design, Evaluation Rubrics, Automated Quality Assessment, Quality Metrics, Data Auditing, Data Quality Validation, Root Cause Analysis, LLM Output Validation, Model Evaluation (Precision, Recall, Cross-Validation)
- **Backend Development:** Flask, ASP.NET Core, Spring Boot, REST APIs, Microservices, Asynchronous Processing, HMAC/JWT Authentication
- **Database & Data Management:** SQL Server, MongoDB, PostgreSQL(pgvector), MySQL, Azure Data Factory, Chroma, ETL Pipelines, NoSQL
- **DevOps & Cloud Services:** AWS (EC2, ECS, Lambda, S3, SageMaker), Azure (VMs, Load Balancer, Key Vault, Monitor, Backup, Site Recovery, ARM Templates), Docker, Kubernetes, CI/CD (GitHub Actions, Azure DevOps)
- **Development Tools & Methodologies:** Git, GitHub, Postman, Swagger, JUnit, Code Reviews, Technical Documentation, Agile/Scrum

EDUCATION

California State University, Fullerton **Aug 2022 - May 2024**
Master of Science, Computer Science | GPA: 3.7/4.0 *Fullerton, CA*

Sreenidhi Institute of Science and Technology **Jun 2017 - May 2021**
Bachelor of Computer Science | GPA: 8.3/10.0 *Hyderabad, India*

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)